HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and developed **Java Spring Boot** REST APIs and backend services for enterprise applications, handling over **1,500 requests per minute** during peak loads.
- Implemented **JPA** entities and managed database interactions with **Hibernate**, ensuring data integrity and efficient persistence for core business logic.
- Engineered **Python** scripts to automate ETL processes, transforming and loading data from disparate sources into a centralized data warehouse, reducing manual processing time by **40%**.
- Collaborated with a global team to build and maintain the next generation of data platforms, focusing on AI-ready capabilities and self-service analytics.
- Contributed to the development of **AI agents** and **MCP-based integrations** to enable natural language interaction with data products.
- Assisted in building and maintaining scalable **data pipelines** and **cloud-native data services** on Azure, ensuring data availability and reliability.
- Containerized services using **Docker** and deployed them to **Azure Kubernetes Service (AKS)**, managing configurations and resource allocation for improved deployment efficiency.

SKILLS

AI/ML Platforms – AI Agents, Agentic Workflows, Model Context Protocol (MCP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Databases, Semantic Search

Backend Development – Java, Spring Boot, Python, REST APIs, gRPC, Event-Driven Architectures

Data Engineering & Processing – Apache Spark, Dremio, Azure Data Factory (ADF), Kafka, Data Pipelines, ETL

Cloud Platforms – Azure Data Platform, Azure Databricks, Azure Kubernetes Service (AKS), Azure Data Lake Storage (ADLS)

DevOps & Tools – GitHub, GitHub Actions, Jira, Confluence, Dynatrace, Datadog, Docker

PROJECTS

AI-Powered Data Discovery Service (HCL Technologies)

- Developed a **Python**-based service with **FastAPI** to expose enterprise data assets, enabling users to discover datasets through a natural language interface.
- Integrated with **PostgreSQL** to retrieve metadata and context for datasets, powering a rudimentary search functionality that improved data discovery time by **15%**.
- Implemented basic prompt engineering techniques to refine user queries before passing them to a simulated knowledge base, enhancing the relevance of search results.

Scalable Data Pipeline for Financial Metrics (HCL Technologies)

- Engineered a **Python** data pipeline using **pandas** and **SQL** to ingest, transform, and load financial transaction data from multiple sources into a central repository.
- Automated the generation of daily operational reports, providing key metrics on transaction volumes and processing status to business stakeholders.
- Deployed the pipeline on **AWS Lambda**, ensuring efficient processing of approximately **2,000 events per hour** and incorporating logging for monitoring and debugging.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)